

Problem Set 9 - Central Bank Policy

1) a) Draw the AA-DD model for Sweden (Krona) [placing the correctly labeled exchange rate ($E_{Kr/\epsilon}$ or $E_{\epsilon/Kr}$) on the vertical axis and Swedish output (Y_{Kr}) on the other axis].

b) Assume the Denmark Central Bank were to **permanently** reduce the money supply of Swedish Krona. Demonstrate the effect of this decrease in the money supply on the Krona exchange rate and Swedish output **in the short-run** (i.e., hold prices fixed) using the AA-DD model

c) Explain the effect of this permanent decrease in the money supply on each of the following prices **in the short-run**.

- i) the price (in Krona) of a bag of frozen Swedish meatballs for sale in Stockholm, Sweden.
- ii) the price (in euros) of a bag of Swedish meatballs for sale in Paris, France.

d) Assume the Denmark Central Bank were to **temporarily** reduce the money supply of Swedish Krona. Explain whether the change in prices in your answer to (c) would be larger or smaller or the same. You should draw and use the asset model picture (combination of IRP and money market) as part of your explanation.

2) Recall that we discussed how expectations of inflation (i.e., expected future higher prices) can lead to current inflation (i.e., a rise in prices today). Assume that there is an expectation that the Danish Central Bank will (at some point in the near future) permanently increase the money supply.

- a) Explain whether people will expect higher or lower prices in the future given this (expected) change in policy.
- b) Assume that the expectation of (a) leads to a current change in prices consistent with your answer to (a). Draw the AA-DD model for Denmark (Krone) [placing the correctly labeled exchange rate ($E_{Kr/\epsilon}$ or $E_{\epsilon/Kr}$) on the vertical axis and Denmark output (Y_{Kr}) on the other axis]. Then demonstrate the effect of this change in prices on the Krone exchange rate and Danish output using the AA-DD model,
- c) Assume the Danish Central Bank has a mandate to maintain full employment. Given the change in output found in (b), what monetary policy should the Danish Central Bank undertake to get the economy back to full-employment?
- d) In literature a “self-fulfilling prophecy” is a phenomenon where an initially false belief or expectation leads to behaviors that make that expectation come true. In the above example, is the expectation of a future increase in the money supply a “self-fulfilling prophecy”?